Experiment 1

Comparative Study of Low-Code Development versus Traditional Application Development
Application: Insurance Claim Request & Processing Management

Aim

To compare low-code development and traditional application development approaches based on development speed, ease of use, customization, cost, maintenance, scalability and overall suitability for different types of software projects.

Theory

Application development is the process of designing, building, testing and deploying software. Two broad approaches are used. **Low-code development** uses visual designers, drag-and-drop components, templates and built-in automation so that useful applications can be assembled with very little hand-written code. **Traditional development** uses programming languages such as Java, Python, C# or JavaScript and gives the developer complete control over architecture, design and behaviour, at the cost of more time and skill.

For this lab series the same business problem is used throughout: an **Insurance Claim Request & Processing Management** application in which policyholders submit claims, and the claims team tracks, verifies and settles them. In Zoho Creator this application is built from a Claim_Request_Form, a Claim_Request_Report, a Claims Command Center page and workflows; in a traditional stack the same system would need a database schema, back-end APIs, a front-end, authentication and hosting to be built separately.

Comparison

Parameter	Low-Code (Zoho Creator)	Traditional Development
Development time	Faster – forms and reports are generated visually	Slower – every layer is coded
Coding requirement	Minimal (Deluge script only where needed)	Extensive
Learning curve	Easy	Moderate to difficult
Customization	Limited to platform components	Highly flexible
Cost	Lower for small and medium projects	Higher initially
Scalability	Moderate (plan limits apply)	High
Maintenance	Easier – platform handles hosting and updates	Needs skilled developers
Insurance claim example	Claim form, status board and reports built in a few days	Custom web app, API and database built over weeks
Best suited for	Simple to medium business applications	Complex enterprise systems

Procedure

1. Study the concepts of low-code and traditional application development.
2. Select a low-code platform (Zoho Creator) and review its drag-and-drop interface, templates and automation tools.
3. Study the traditional development process using programming languages and frameworks.
4. Compare both approaches on development time, coding effort, customization, scalability, cost, maintenance and deployment.
5. Record the observations in a comparison table and rate each parameter.
6. Analyse advantages and limitations of each approach.
7. Conclude which approach suits which type of project, using the insurance claim application as the case study.

Output

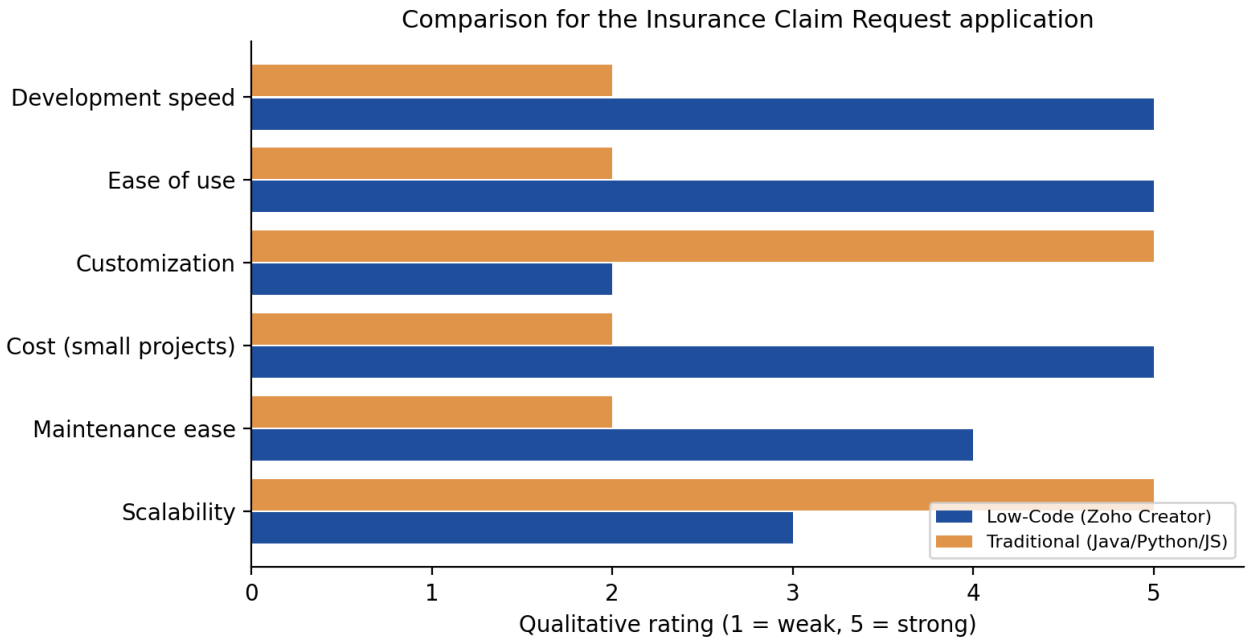


Figure 1.1 – Qualitative comparison (ratings are the author's assessment, not measured data)

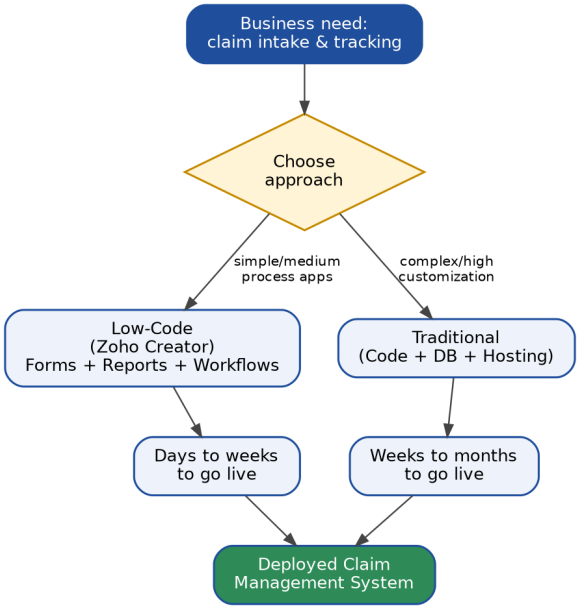


Figure 1.2 – Choosing an approach for the claim management application

Advantages and Limitations

- **Low-code advantages:** rapid delivery, lower cost, built-in security and hosting, usable by citizen developers.
- **Low-code limitations:** restricted UI customization, plan-based limits, dependence on the vendor platform.
- **Traditional advantages:** full control, unlimited customization, suited to large complex systems.
- **Traditional limitations:** long development time, higher cost, need for specialised developers.

Result

The comparative study showed that low-code development (Zoho Creator) is faster, cheaper and easier to maintain for simple to medium business applications such as the Insurance Claim Request system, whereas traditional development is better for complex, highly customised and large-scale systems.